



HabitMiner: A Context-Aware Framework for Personalized Routine Discovery Using Mobile Sensor Data

PERVASIVE COMPUTING • GROUP NO. [8]

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Problem Statement

- Existing digital wellness applications provide aggregate statistics such as screen time and app usage, but fail to understand an individual's normal digital routines.
- This project aims to mine personalized smartphone usage habits and detect meaningful deviations from a user's own behavioural baseline using context-aware analysis.

WHAT TODAY'S WELLNESS TOOLS REPORT

Screen time	4 h 12 m
Instagram	1 h 30 m
YouTube	48 m
Device unlocks	84
Notifications	213

Accurate — and silent on whether any of it was normal for this person.



WHAT HABITMINER REPORTS

Unusual night · deviation 0.79

Social media ran 02:14–03:58 — 104 minutes, against a normal 6 minutes after 23:00.
The usual 20:47 Instagram → YouTube routine did not fire.
Main driver: late-night volume, 38% of the score.

Describes the change relative to this person's own routine, and names why.

WHY THIS IS HARD

No population norm

What is normal for one person is unusual for another, so averages hide the individual.

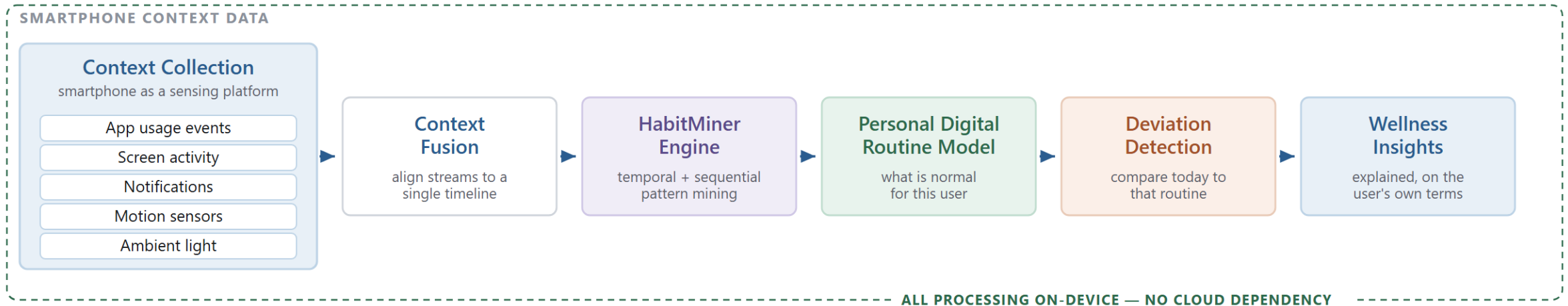
Context changes meaning

The same app at the same hour is a different behaviour when walking than when still in the dark.

No labels for “unusual”

No dataset marks which days were atypical, so the baseline has to be learned from the user.

Methodology — System Overview



WHAT EACH SIGNAL CONTRIBUTES

— *physical context disambiguates identical digital behaviour*

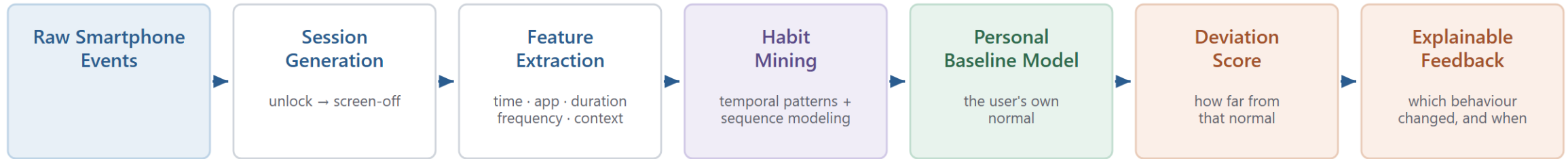
App usage events	Screen activity	Notifications	Motion sensors	Ambient light
which activity the user chose	when it happened and for how long	what prompted the interaction	still, walking or travelling	day or night, indoors or outdoors

- The smartphone acts as a pervasive sensing platform, capturing both digital interaction and physical context as the user goes about an ordinary day.
- Multimodal context is collected and processed locally on the device, with no cloud dependency and no raw data leaving the handset.
- Motion and light readings disambiguate identical digital behaviour, so the model learns situations rather than clock times alone.

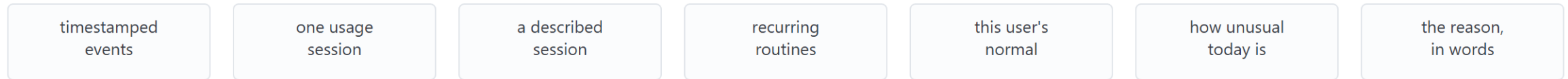
Methodology — Processing Pipeline



PROCESSING PIPELINE



WHAT EACH STAGE PRODUCES



Each stage is inspectable — nothing is a black box, and no raw event ever leaves the device.

- Recurring digital routines are identified using temporal and sequential pattern mining over the user's own session history.
- A personal baseline is built for each user, instead of comparing that user against population averages.
- Deviations are explained by identifying which behaviour changed and in which part of the day, rather than reporting an unexplained score.

Evaluation — Datasets and Baselines



DATASETS

StudentLife

PUBLIC

Smartphone sensing and behavioural traces from university students, collected over a 10-week term.

Used for: Digital behaviour and context analysis · Screen sessions and sensor context

UCI Human Activity Recognition

PUBLIC

Accelerometer and gyroscope recordings with labelled physical activities.

Used for: Physical context recognition · Motion features for still / walking

Android prototype traces

SELF-COLLECTED

Usage and sensor logs generated by the team's own devices during the trial period.

Used for: Prototype validation · App usage, screen sessions, motion and light

BASELINES FOR COMPARISON

1

Statistical usage baseline

Daily averages and fixed thresholds.

Misses: Cannot say when a change happened or which behaviour caused it.

2

Frequency-based prediction

Most frequently occurring behaviour.

Misses: Ignores context — predicts the same app whatever the situation.

3

HabitMiner (proposed)

OURS

Context-aware personalized routine model.

Adds: Learns one person's routine and explains what changed.

EVALUATION PROTOCOL

Models are trained on each user's earlier days and tested on their later days; data is never shuffled.
Prototype traces validate the system on current, real-world usage from the team's own devices.

Public datasets support offline development; the Android prototype supplies live traces for validation on real, present-day usage.

Evaluation — Metrics and Expected Results



METRICS

Habit discovery

- Pattern stability across time windows
- Similarity between discovered routines

Next-behaviour prediction

- Top-1 next-behaviour prediction accuracy
- Top-3 next-behaviour prediction accuracy

Deviation detection

- Precision
- Recall
- False positive rate

System performance

- Battery impact per day
- On-device processing latency

EXPECTED RESULTS

Routines beyond raw statistics

Context-aware modelling should identify personal routines that simple usage statistics cannot express.

Fewer false alarms than fixed thresholds

Personalized baselines should flag meaningful behavioural change while raising fewer false alarms than population-wide thresholds.

Practical on-device operation

The prototype should learn routines and score each day within an ordinary battery and latency budget, without offloading any data.

Deviation detection is validated by injecting known changes into held-out traces, and by asking users whether flagged days felt unusual.



Conclusion and Future Work

CONCLUSION

HabitMiner transforms the smartphone from a passive usage tracker into a context-aware system capable of learning a person's own digital routines.

Digital interaction and physical context are fused on the device, so the system captures not only what was used but the situation it was used in.

The result is privacy-preserving and explainable digital wellness feedback, with all analysis performed locally.

FUTURE WORK

Long-term personalized routine learning across months rather than weeks.

Federated learning for privacy-preserving model sharing across users.

More advanced on-device activity recognition.

Adaptive personalized interventions built on the learned routine model.

EXPECTED DELIVERABLES

Android prototype application

on-device collection, mining and dashboard

Evaluation on public data + live traces

StudentLife, UCI HAR and team-collected usage

Explainable wellness insights

routine summaries and reasons for flagged days

SCOPE OF THIS PROJECT

HabitMiner describes behaviour and explains change. It does not diagnose, and it does not block or restrict app usage.